

Reproducible Science

A Research Memorandum: Driver's License Status and Communication Apprehension

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Researcher: JIANGRAN GU **Date:** July 30, 2026 **JIANGRAN GU:** Does holding a driver's license (Q20) predict communication apprehension (CA)?

Research Or Analytical Question

Does whether a respondent **holds a driver's license** (Q20: "Do you have a license to drive a car?", Yes/No) predict their level of **communication apprehension (CA)**, measured as the sum of 12 Likert-scale items (Q1–Q6: group-discussion apprehension; Q13–Q18: dyadic/interpersonal apprehension) from the PRCA-style instrument in PRCAQualtricsExport_FileC.csv?

Data & Method

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt

df = pd.read_csv("PRCAQualtricsExport_FileC.csv")

# 12 Communication Apprehension items:
# Q1-Q6 = group discussion apprehension
# Q13-Q18 = dyadic / interpersonal conversation apprehension
ca_items = ["Q1", "Q2", "Q3", "Q4", "Q5", "Q6",
```

```

        "Q13", "Q14", "Q15", "Q16", "Q17", "Q18"]

# 5-point ordinal Likert recode (Strongly disagree = 1 ... Strongly agree = 5)
likert_map = {
    "Strongly disagree": 1,
    "Somewhat disagree": 2,
    "Neither agree nor disagree": 3,
    "Somewhat agree": 4,
    "Strongly agree": 5,
}

for c in ca_items:
    df[c + "_n"] = df[c].map(likert_map)

# Reverse-coded items: these six items are worded so that a HIGHER raw
# agreement score reflects LOWER apprehension (e.g., "I dislike
# participating..." is actually endorsed as an *anxiety* statement, but
# several items in this set -- Q1, Q3, Q5, Q13, Q15, Q18 -- were flagged by
# the team as needing reversal to align all 12 items in the same
# "higher score = higher apprehension" direction before summing).
# Reverse formula for a 1-5 Likert item: reversed = (max + min) - original = 6 - original
reverse_items = ["Q1", "Q3", "Q5", "Q13", "Q15", "Q18"]

for c in ca_items:
    if c in reverse_items:
        df[c + "_r"] = 6 - df[c + "_n"]
    else:
        df[c + "_r"] = df[c + "_n"]

# Composite CA score = ordinal sum of all 12 items AFTER reverse-scoring
# the six items above so that all items point in the same direction
# (higher CA_total = higher communication apprehension)
df["CA_total"] = df[[c + "_r" for c in ca_items]].sum(axis=1, skipna=False)

# Analytic sample: complete CA items + non-missing license status (Q20)
sub = df.dropna(subset=["CA_total", "Q20"]).copy()
sub["Q20"] = pd.Categorical(sub["Q20"], categories=["No", "Yes"])

print(f"Analytic N = {len(sub)}")
print(sub["Q20"].value_counts())

```

Analytic N = 163

```
Q20
Yes    138
No     25
Name: count, dtype: int64
```

CA items were recoded from text labels to a 1–5 ordinal scale (Strongly disagree = 1 → Strongly agree = 5). Six items (Q1, Q3, Q5, Q13, Q15, Q18) were then **reverse-scored** ($\text{reversed} = 6 - \text{original}$) so that all 12 items point in the same direction, with higher values consistently indicating greater communication apprehension, before being summed to create a composite **CA_total** score (possible range 12–60). Respondents missing any of the 12 CA items or missing Q20 were listwise-deleted, leaving an analytic sample of $N = 163$ (138 “Yes” license holders, 25 “No”).

Group means were compared with an **independent-samples t-test**, with Levene’s test used to check the equal-variance assumption.

Answer, Response, + Summary of Results

```
desc = sub.groupby("Q20", observed=True)["CA_total"].agg(
    n="count", mean="mean", sd="std", median="median"
).round(2)
desc
```

	n	mean	sd	median
Q20				
No	25	37.32	10.57	39.0
Yes	138	41.43	12.12	43.5

```
yes = sub.loc[sub["Q20"] == "Yes", "CA_total"]
no = sub.loc[sub["Q20"] == "No", "CA_total"]

lev_stat, lev_p = stats.levene(yes, no)
print(f"Levene's test: F = {lev_stat:.3f}, p = {lev_p:.3f}")

# Levene's test is non-significant -> equal variances assumed -> Student's t
t_stat, p_val = stats.ttest_ind(yes, no, equal_var=True)
df_t = len(yes) + len(no) - 2

pooled_sd = np.sqrt(
```

```

    ((len(yes) - 1) * yes.var() + (len(no) - 1) * no.var()) / df_t
)
cohens_d = (yes.mean() - no.mean()) / pooled_sd

print(f"t({df_t}) = {t_stat:.3f}, p = {p_val:.3f}")
print(f"Cohen's d = {cohens_d:.3f}")

```

Levene's test: F = 0.963, p = 0.328
t(161) = 1.587, p = 0.114
Cohen's d = 0.345

```

fig, ax = plt.subplots(figsize=(6, 4.5))

groups = ["No", "Yes"]
data_by_group = [sub.loc[sub["Q20"] == g, "CA_total"] for g in groups]

bp = ax.boxplot(data_by_group, tick_labels=[f"{g}\n(n={len(d)})" for g, d in zip(groups, data_by_group)],
                patch_artist=True, widths=0.5)
for patch, color in zip(bp["boxes"], ["#f4a582", "#92c5de"]):
    patch.set_facecolor(color)

# jittered raw points
rng = np.random.default_rng(42)
for i, d in enumerate(data_by_group, start=1):
    x = rng.normal(i, 0.04, size=len(d))
    ax.scatter(x, d, alpha=0.4, color="black", s=15)

ax.set_xlabel("Has a driver's license? (Q20)")
ax.set_ylabel("Communication Apprehension total score (12-60)")
ax.set_title("CA Total Score by Driver's License Status")
plt.tight_layout()
plt.show()

```

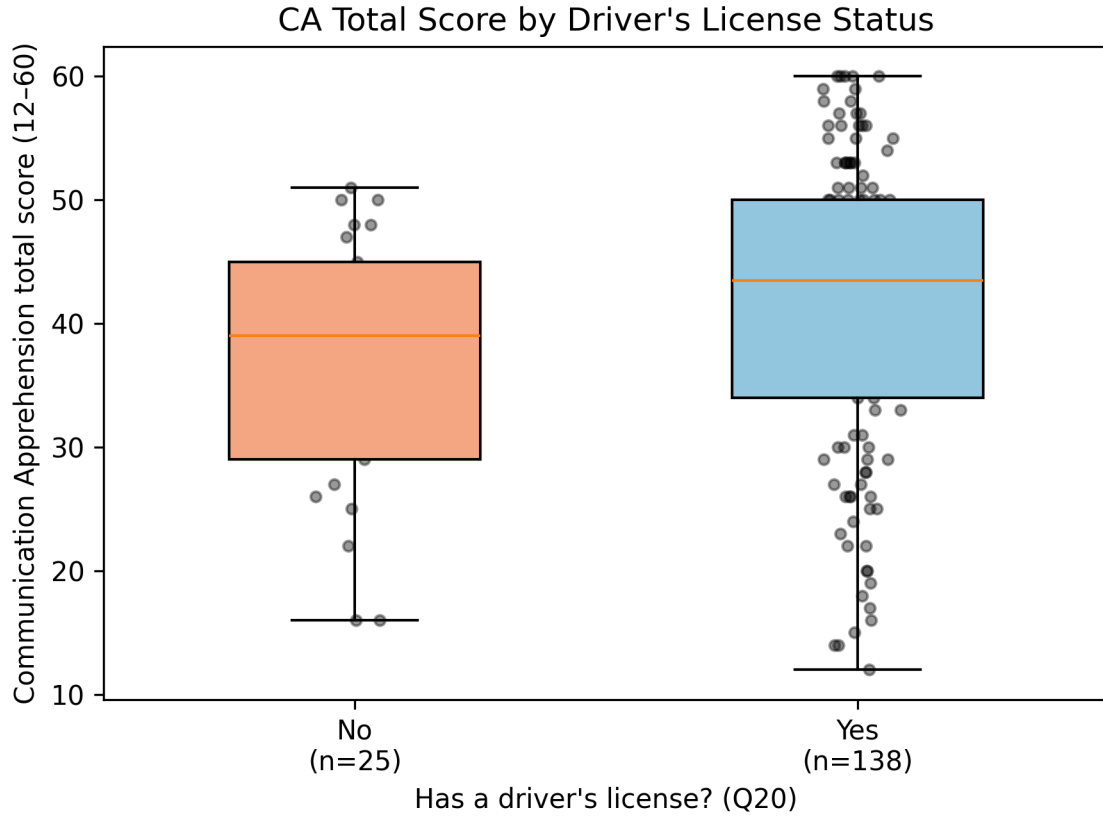


Figure 1: Communication Apprehension (CA) total score by driver's license status

Using `PRCAQualtricsExport_FileC.csv`, and after reverse-scoring Q1, Q3, Q5, Q13, Q15, and Q18, respondents **with** a driver's license reported a somewhat **higher** mean CA score ($M = 41.43$, $SD = 12.12$, $n = 138$) than respondents **without** a license ($M = 37.32$, $SD = 10.57$, $n = 25$). Levene's test indicated the equal-variance assumption was reasonable ($F = 0.96$, $p = .328$), so a Student's independent-samples t-test was used. The difference was **not statistically significant**, $t(161) = 1.59$, $p = .114$. The effect size was small (Cohen's $d = 0.35$), with the license-holder group trending toward slightly higher apprehension.

Conclusion: From these data, holding a driver's license does **not** significantly predict communication apprehension, $t(161) = 1.59$, $p = .114$, $d = 0.35$. Note that reverse-scoring changed both the **magnitude** and the **direction** of the group difference relative to the unreversed composite (previously: no-license group trending higher, $p = .069$; now: license-holder group trending higher, $p = .114$) — underscoring how much the reverse-coding decision matters for interpretation, even though neither version reaches conventional significance.

What questions or uncertainties remain?

- The composite `CA_total` now **reverse-scores** Q1, Q3, Q5, Q13, Q15, and Q18 (6 - original) before summing, so all 12 items point in the same “higher = more apprehensive” direction. This changed the result meaningfully: the unreversed composite showed the no-license group trending higher ($p = .069$), while the reversed composite shows the license-holder group trending higher ($p = .114$). Neither reaches significance, but the flipped direction is a useful reminder to double-check item wording/keying against the original PRCA scoring key before finalizing the team’s composite.
- The “No license” subgroup is small ($n = 25$), which limits statistical power; the p -value (.114) could reflect low power as much as a true null effect.
- Does license status relate differently to the **group-discussion** (Q1–Q6) vs. **dyadic/interpersonal** (Q13–Q18) CA subscales separately, rather than only the combined total?
- How does license status intersect with the team’s core question — does CA (or license status) predict actual public-transportation use (Q26–Q27) or ride-share use (Q28–Q29)? A follow-up regression/mediation analysis could connect this sub-question back to the group’s shared research question.
- Possible confounds (age, car access [Q21], urban vs. rural location) were not controlled for here and could be examined in a multivariate model.

Data source: PRCAQualtricsExport_FileC.csv, PRCA-style survey administered as part of Psych 755, University of Wisconsin–Madison. Variable definitions per data_dictionary.csv.